Negative Binommial Regression Article

1 Background

1.1 Understanding the Technique

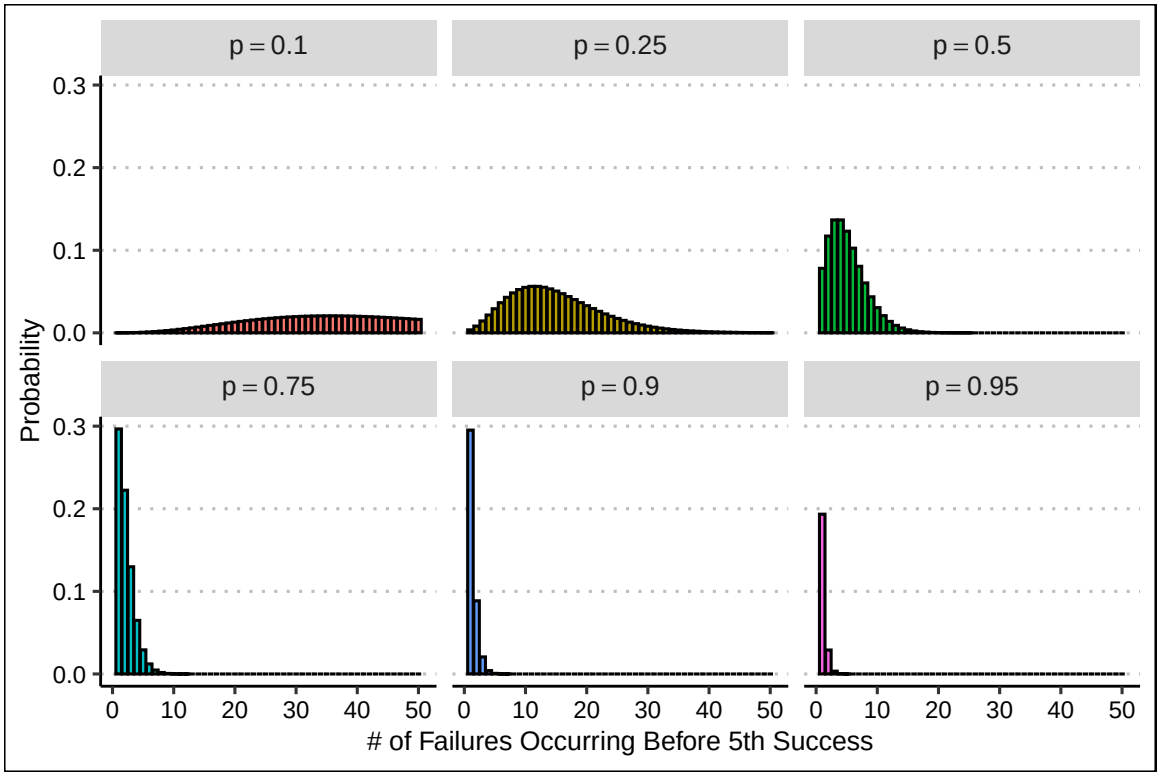
1.1.1 The Negative Binomial Distribution

Negative binomial regression extends the linear least-squares regression model to data that fits a negative binomial distribution. My wording there is careful, because most often the data doesn't fit what the negative binomial is actually modeled after. The negative binomial distribution measures the probability of X failures occuring before the n-th success, where the probability of a success is p. For example, the probability of getting $X = 6$ tails before getting the $n = 3$ heads, where the the probability of getting a head or tail $p = 0.5$.

$$P(X = x|n, p) = f_x(x, n, p) = \binom{x + n - 1}{n - 1} p^n (1 - p)^x$$

We can plot the probability mass function (PMF) to see the probability of x failures occuring for every X, while changing the probability of success and keeping our number of success targets constant:

Figure 1: Negative Binomial Distribution Examples



To set up the regression model, we need to work out the mean and variance of our distribution (statisticsmatt 2020).

Expected Value / Mean:

$$
\begin{aligned}
E(X) =& \sum_{x=0}^{\infty} x \cdot f_X(x,n,p) \\
=& \sum_{x=0}^{\infty} x \cdot \binom{n+x-1}{n-1} p^n (1-p)^x \\
=& np^{-1}(1-p) \cdot \sum_{x=1}^{\infty} \frac{(n+x-1)}{n!(x-1)!} p^{n+1}(1-p)^{x-1} \\
(\text{Let } y = x-1) =& np^{-1}(1-p) \cdot \sum_{y=0}^{\infty} \frac{(n+y)!}{n!\,y!} p^{n+1}(1-p)^{y} \\
=& \frac{n(1-p)}{p} \cdot \sum_{y=0}^{\infty} \binom{(n+1)+y-1}{y} p^{n+1}(1-p)^{y} \\
=& \frac{n(1-p)}{p}
\end{aligned}
$$

The second moment of the negative binomial distribution is given by:

$$
\begin{aligned}
E[X(X-1)] =& \sum_{x=0}^{\infty} x(x-1) f_x(x) = \sum_{x=0}^{\infty} x(x-1) \frac{(n+x-1)!}{(n-1!x!} p^n (1-p)^x \\
=& \frac{n(n+1)(1-p)^2}{p^2} \cdot \sum_{x=2}^{\infty} \frac{[(n+2)+(x-2)-1]!}{(n+1)!(x-2)!} p^{n+2}(1-p)^{x-2} \\
(\text{Let } y = x-2) =& \frac{n(n+1)(1-p)^2}{p^2} \cdot \sum_{y=0}^{\infty} \binom{(n+2)+y-1}{y} p^{n+2}(1-p)^{y} \\
=& \frac{n(n+1)(1-p)^2}{p^2}
\end{aligned}
$$

To solve for the variance:

$$
\begin{aligned}
Var(x) &= E(x^2) - E(x)^2 = E(x^2) - E(x) + E(x) - E(x)^2 \\
&= E[x(x-1)] + E(x) - E(x)^2 \\
&= \frac{n(n+1)(1-p)^2}{p^2} + \frac{n(1-p)}{p} - \frac{n^2(1-p)^2}{p^2} \\
&= \frac{n^2(1-p)^2 + n(1-p)^2 + np(1-p) - n^2(1-p)^2}{p^2} \\
&= \frac{n(1-p)[(1-p)+p]}{p^2} \\
&= \frac{n(1-p)}{p^2}
\end{aligned}
$$

We can verify this numerically. Here I simulate 10,000 negative binomial distributed datasets with $n = 20$ successes waiting for, probability $p = 0.75$ and sample size $N = 25$:

```
set.seed(13)
sims = 10000 # Simulations
n = 20  # Successes
N = 25  # Sample Size
p = 0.75 # Probability
```

```r

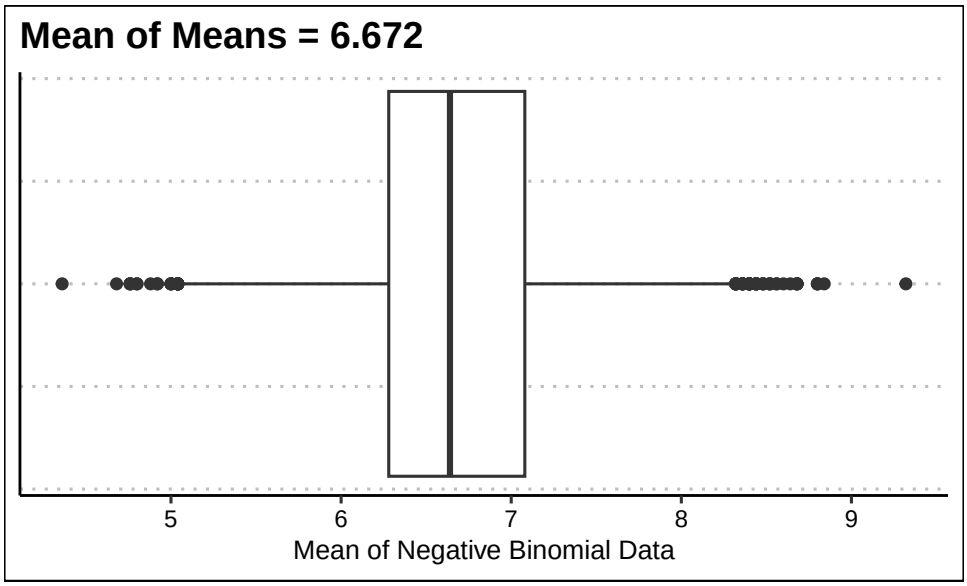
means <- rep(NA, sims) #Place to store means
vars <- rep(NA, sims)  #Place to store variances

for(i in 1:sims){
  nbinom_dat <- rnbinom(n = N, size = n, prob = p) #Simulate random data
  means[i] <- mean(nbinom_dat)   #Store the mean of the data
  vars[i] <- var(nbinom_dat)     #Store the variance of the data
}
```

Theoretically we'd expect our mean number of failures to be:

$$\frac{n(1-p)}{p} = \frac{20(1-0.75)}{0.75} = 6.6\bar{6}$$

Figure 2: Negative Binomial Mean



And our variance in the number of failures to be:

$$\frac{n(1-p)}{p^2} = \frac{20(1-0.75)}{(0.75)^2} = 8.8\bar{8}$$

Note that the boxplot shows the median of all the datasets.

For some context, the variance of the negative binomial distribution approaches the variance of a poisson distribution as the number of failures increases while keeping the mean and successes the same:

1.1.2 Negative Binomial Regression

Where the typical linear regression goes as follows:

$$Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_{p-1} x_{(p-1)i} + \epsilon_i$$

Since Y_i is assumed to be continuous, both sides of the equation can range from $(-\infty, \infty)$. To model a variable that is based on counts, and therefore strictly > 0, we have to transform the data to sit in the $(-\infty, \infty)$ range, and this can be achieved with a log transformation:

Figure 3: Negative Binomial Variances

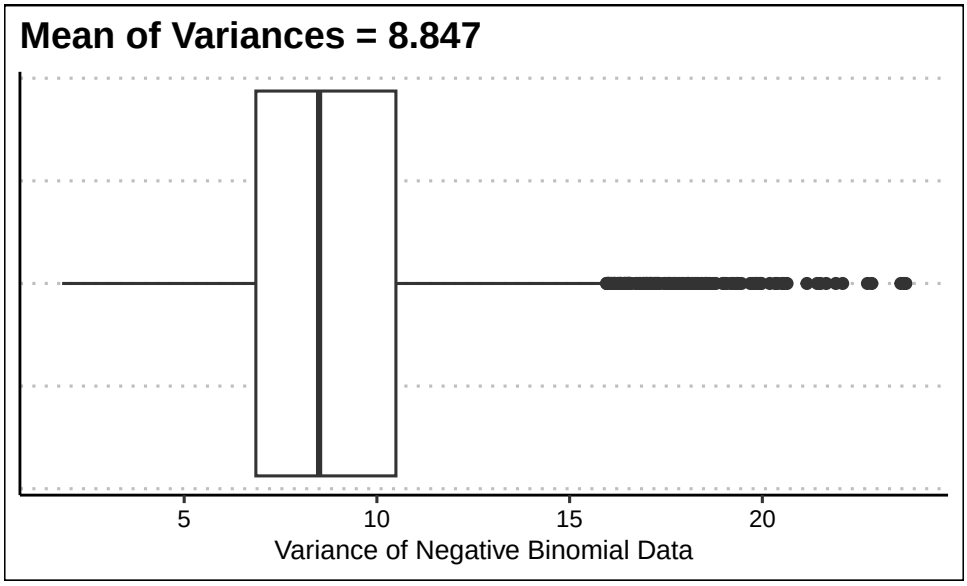
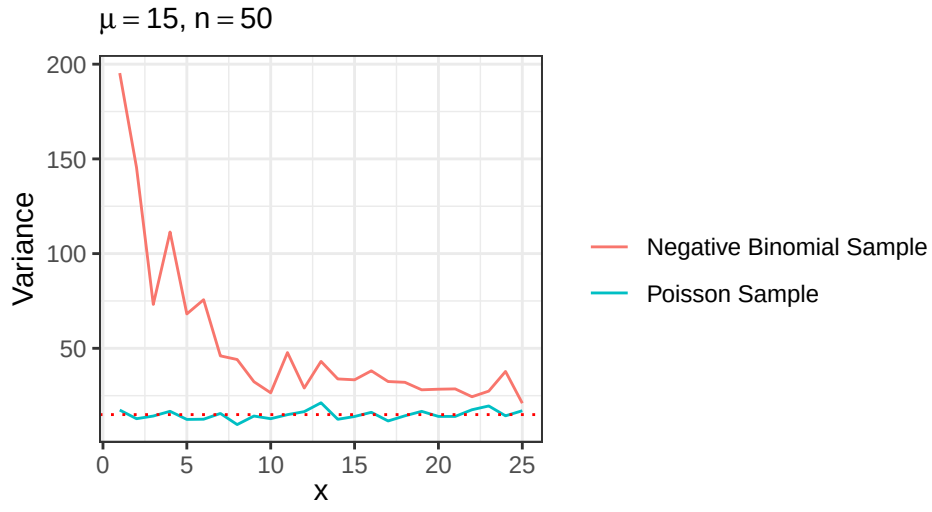


Figure 4: Negative Binomial and Poisson Variances



$$log(\mu_{Y_i|x_i}) = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_{p-1} x_{(p-1)i}$$

where $\mu_{Y_i|x_i}$ is the mean of the Negative Binomial distribution from which the ith response is drawn.

1.2 The Study

Figure 5: News-Medical.net



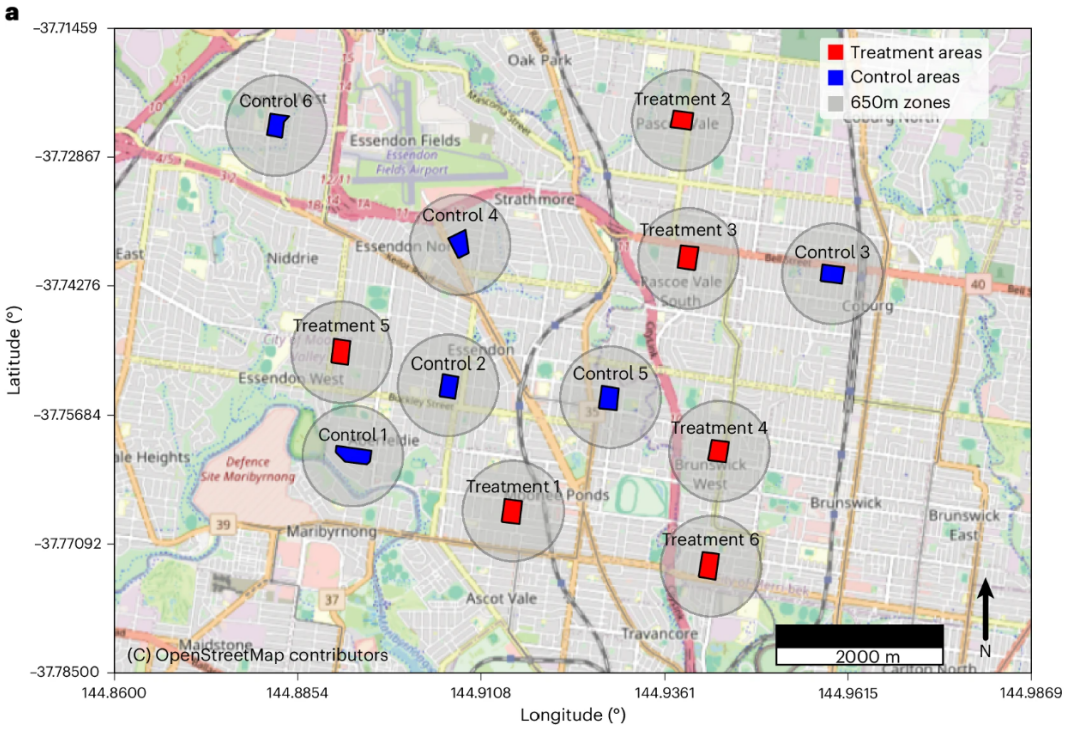
Everybody hates mosquitoes, especially Australians. *Aedes notoscriptus* mosquitoes, also known as the Australian backyard mosquito, are particularly harmful in any area where they're found. These mosquitoes have been found to spread *Mycobacterium ulcerans* in humans and other animals, resulting in a severe skin condition known as Buruli ulcer (no, I'm not showing a picture for that). In areas around Melbourne, cases of infection have been increasing in recent years, affecting potentially hundreds or thousands of people. People are often exposed in the warmer months where the mosquitoes are active, but because of a 3-6 month incubation period, can have symptoms much later.

Figure 6: Paris et al.



In2Care™ Mosquito Stations attempt to reduce adult populations and prevent the development of immature mosquitoes. Luring the adults in with scented water, the laid eggs and the adults are in contact with pyriproxyfen, which prevents the eggs' normal development and growth, and the adults spread to other areas. The adult mosquitoes are also infected with *Beauveria bassiana*, a fungus that is deadly to them eventually.

Figure 7: Paris et al.



So for this study, treatment and control sites were placed across a 50 km area in the suburbs of Melbourne, Australia,

with each area having 80 partipating residential houses and 100 mosquito dissemination stations each. Baseline levels of mosquito populations were taken for 3 weeks prior to the intervention, with counts taken 6 weeks later for an additional 4 weeks (Paris et al. 2026).

1.3 Goal of Statistical Analysis

Because I can't seem to escape mixed effect models, the negative binomial regression predicts mosquito eggs with factors:

- Treatment Status
- Sampling
- Treatment Status x Sampling Interaction
- Trap Site (Random Intercept)

In *R*, our formula call looks like this:

```
formula <- eggs ~ group*week + (1|site)
```

2 Reproduction of the Study

2.1 Data Cleaning

Data is very nice! I removed the rows with NAs for egg counts as I'm assuming data is unavailable for those collections.

```
dat <- read_csv('mosquitos.csv')

dat$site <- factor(dat$site, levels = c("1", "2", "3", "4", "5", "6", "7", "8", "9", "10", "11", "12"))
dat$group <- factor(dat$group, levels = c("control", "treatment"))

dat <- dat |>
  drop_na()
```

2.2 Data Summary

We can look at each study site's mosquito eggs over time:

Let's take a quick summary of our mosquito egg counts by treatment group and week:

This looks promising for the effectiveness of the treament! Interestingly, the egg counts for both the control and treatment sites drops sharply after the intervention, indicating that the effects of the intervention could be having a crossover effect into the control areas.

2.3 Model Assumptions

2.3.1 Gaussian Residuals

Since our model is a generalized linear mixed effect model, we need to check the assumption that our random intercepts for our sites are gaussian:

Even with a small sample ($n = 12$ sites), our random intercepts look roughly gaussian and follow the qq-plot pretty well.

2.3.2 Dispersion

Dispertion ratios aren't quite as straightforward for mixed models, so we'll be using this approximation function provided by Dr. Ben Bolker, a mathematical statistician at McMaster University (Bolker et al. 2025).

Our dispertion ratio here is **extremely** low. The randomized quantile residuals don't look terrible though.

We can also check our predictions against the standardized Pearson residuals:

This is a better sign at least. We see a relatively flat line near 1 which is good.

Figure 8: Treatment and Week Across Sites

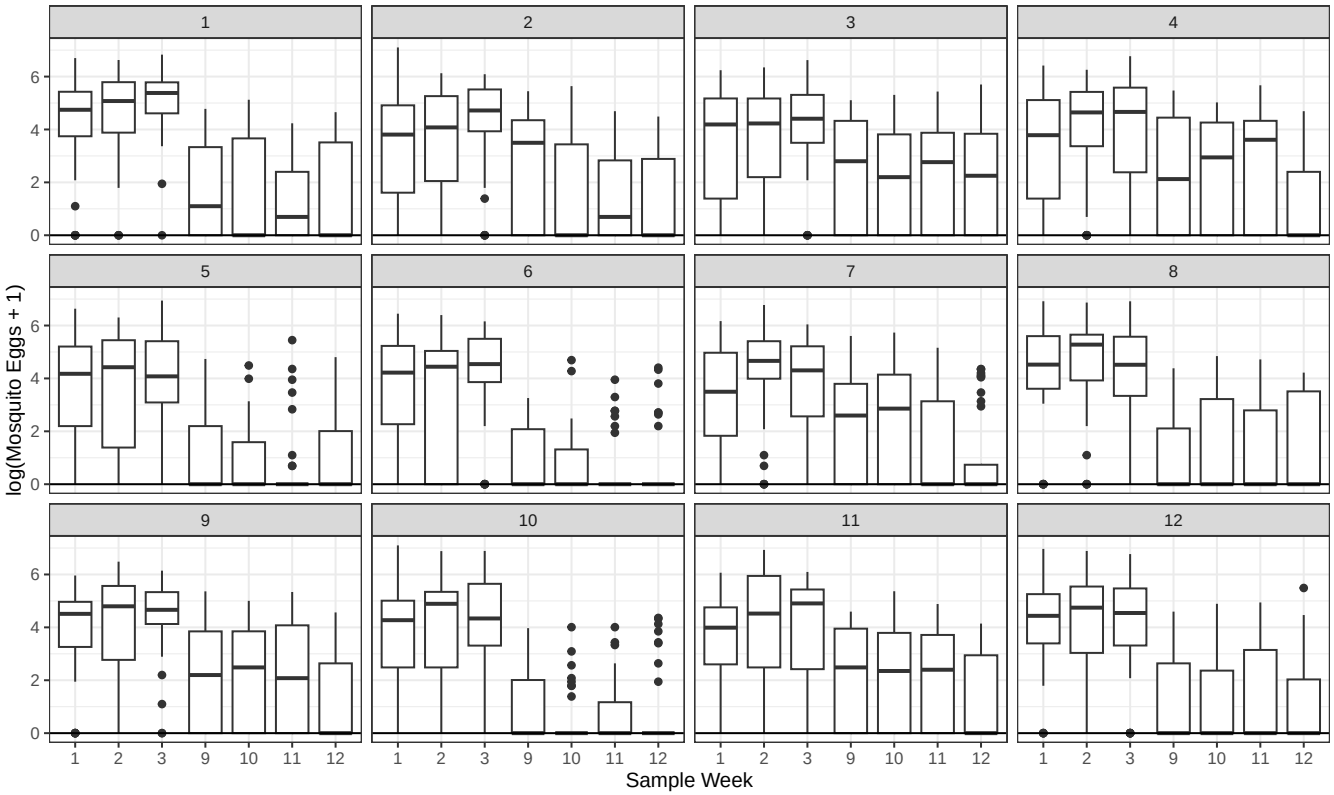


Figure 9: Treatment and Week Summary

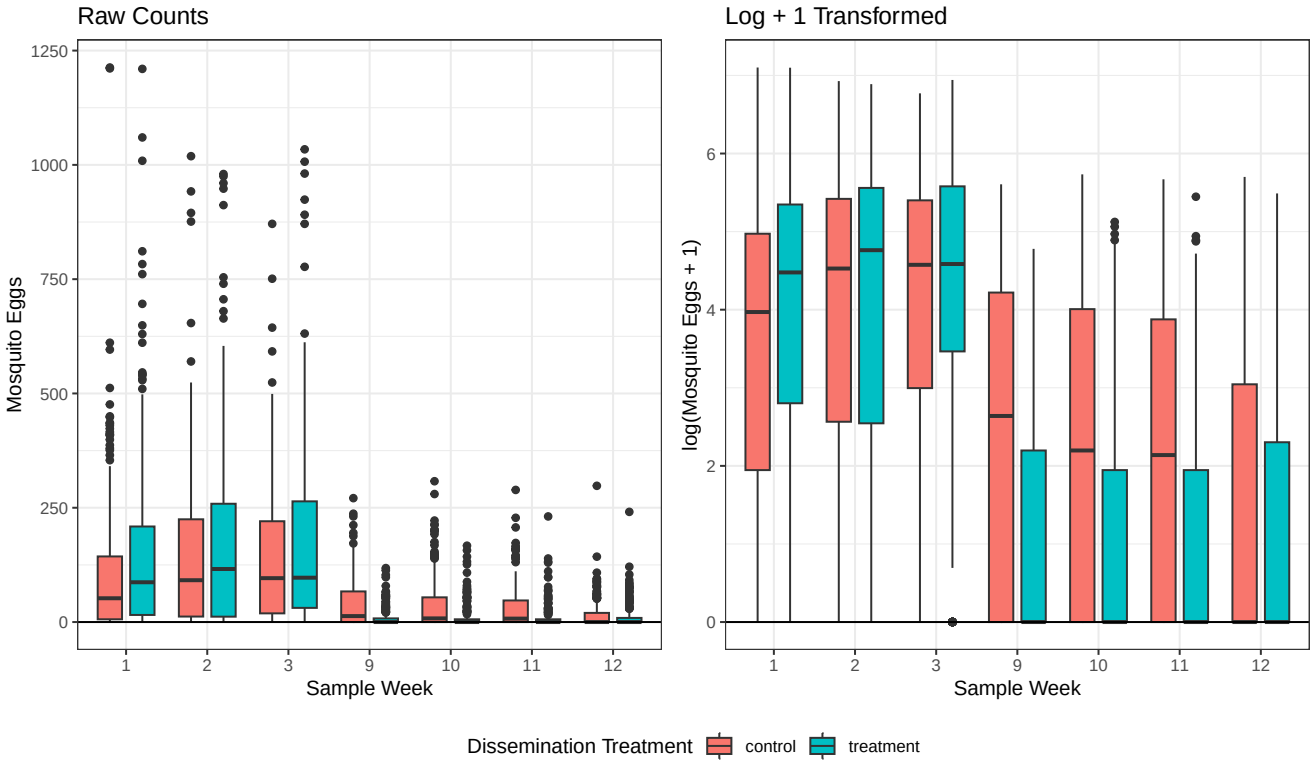


Figure 10: Random intercepts of Sites

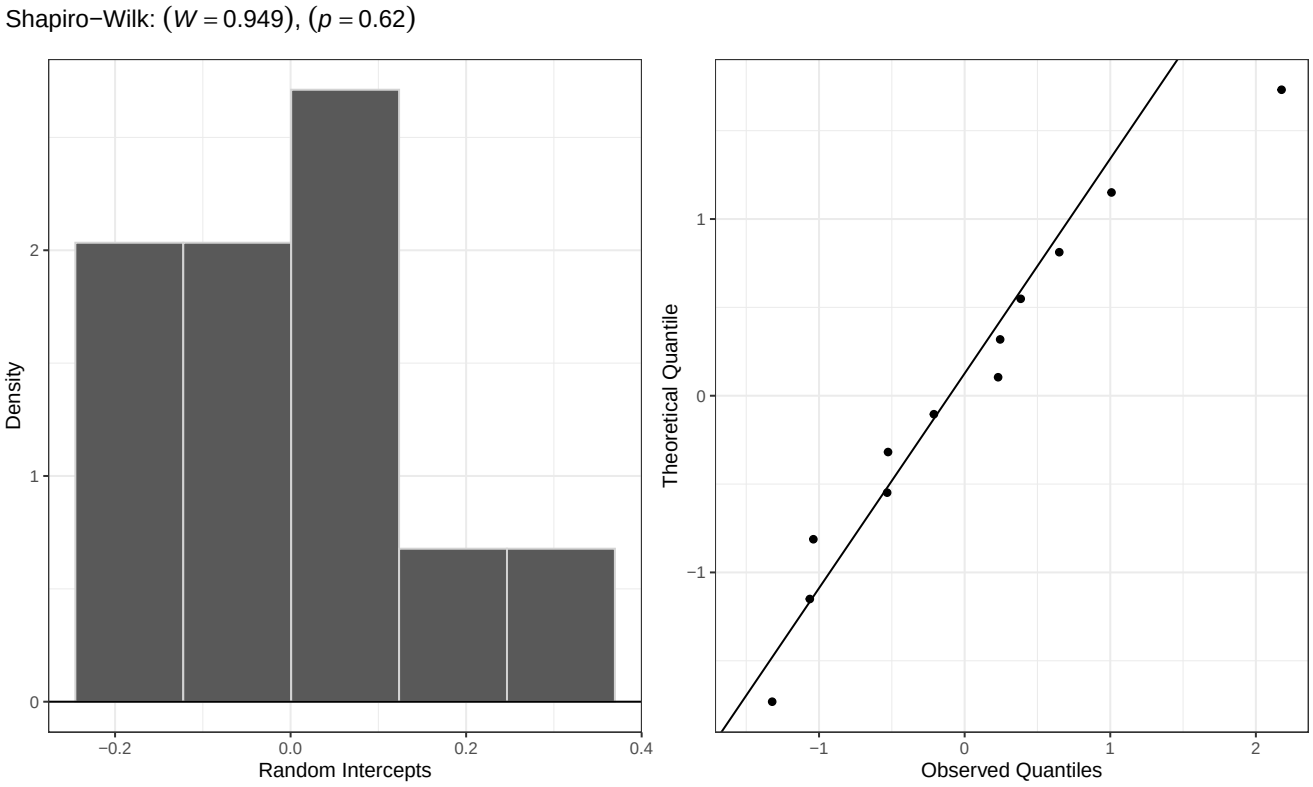


Figure 11: Randomized Quantile Residuals of Negative Binomial Model

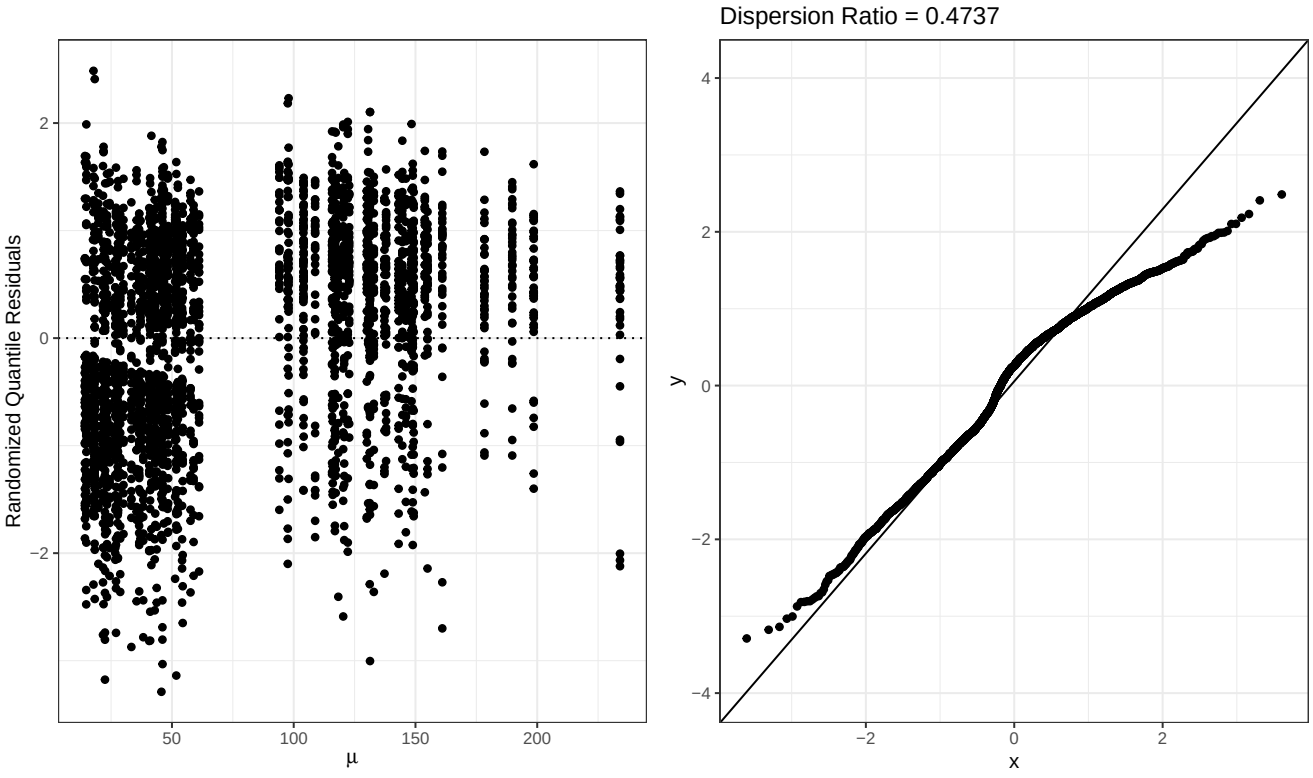
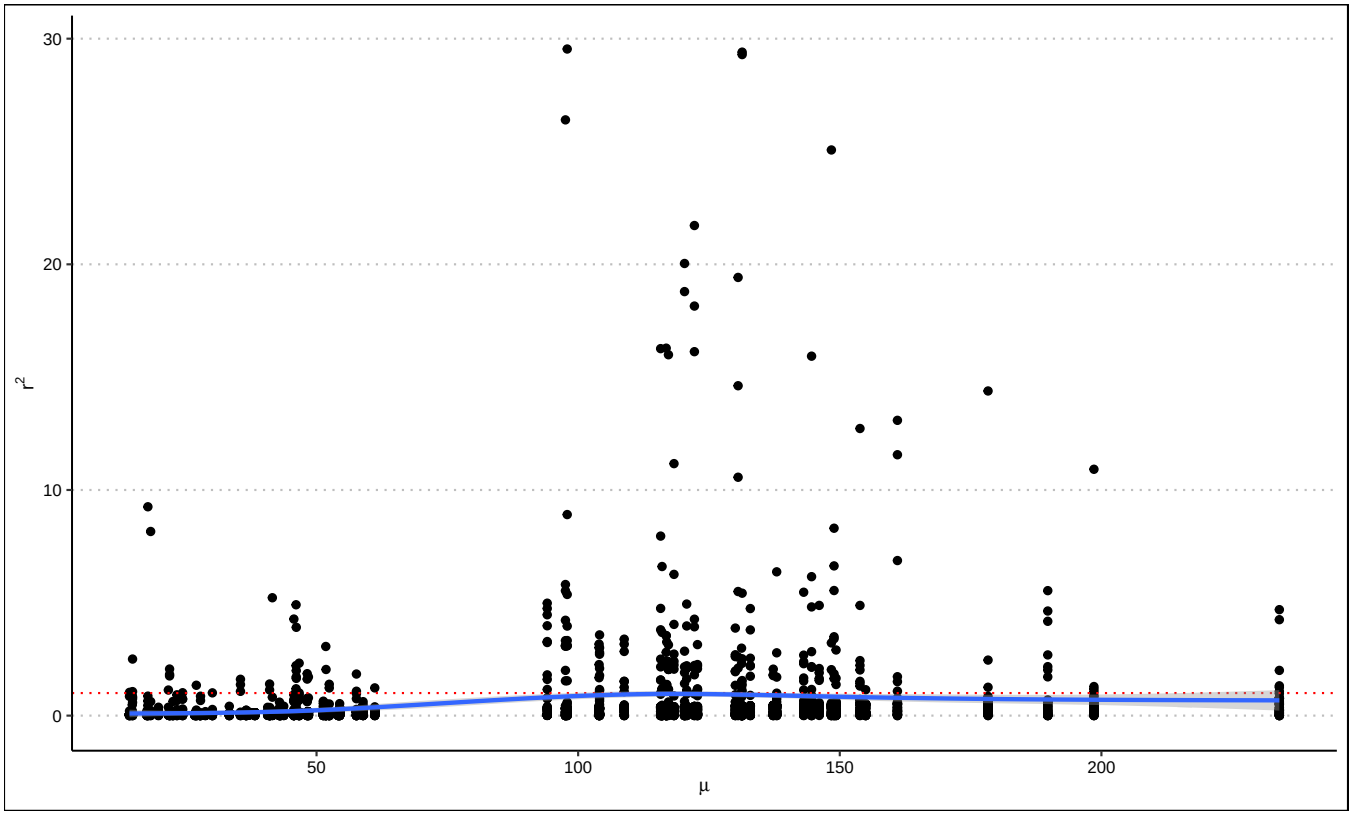


Figure 12: Predictions Against Pearson Residuals



2.3.3 Multicollinearity

Table 1: Model VIF

Term	VIF	Lower CI	Upper CI
Treatment	1.278	1.229	1.337
Week	1.729	1.649	1.819
Treatment : Week	1.951	1.856	2.056

The collinearity for the model looks good, with all the VIFs under 2.

2.3.4 Sufficient Sample Size for Predictors

Ideally we should have at least 10 events counted per predictor:

$$\sum_{i=1}^{n=3,213} \lambda_i = 253,237 \geq 10p = 10(3) = 30 \quad \checkmark$$

2.4 Model Fit

We can obtain an overall model fit:

Table 2: Model Fit

AIC	AICc	BIC	R2_conditional	R2_marginal	ICC	RMSE	Sigma	Score_log	Score_spherical
26712.43	26712.46	26748.88	0.974	0.937	0.586	128.399	301.955	-5.592	0.013

Our fixed effects explain 93.7% of the variance in mosquito eggs (extremely high). The total model including the random effects explains 97.4% of the variance in mosquito eggs.

2.5 Model Inference

Now to see if the treatment on the sites had an effect on the number of mosquito eggs, and if time had any additional effects:

A general summary our model gives us:

Table 3: Model Summary

	Estimate	SE	Z	p	significance	Percent Change
Intercept	5.066	0.081	62.459	<0.001	***	
Group = Treatment	0.291	0.109	2.669	0.008	**	33.8
Week	-0.116	0.007	-17.443	<0.001	***	-11
Group = Treatment : Week	-0.093	0.010	-9.270	<0.001	***	-8.9

where the percent change $= (e^{\text{estimate}} - 1) \times 100$.

Because all of the terms are included in interactions, we'll skip the interpretations for now.

For the intercept:

- Our model predicts that we'd see $e^{5.066} = 158.816$ eggs for a control site in week 0.

Now let's try to untangle all those interactions.

2.6 Model Interactions

2.6.1 Effect of Mosquito Intervention

Testing the contrast of mosquito eggs by treatment group:

Table 4: Contrast of Treatment Groups

Contrast	Estimate	SE	Z	p	Percent Change	Upper CI	Lower CI
control - treatment	0.348	0.098	3.54	<0.001	41.639	71.745	16.81

Plotting the predicted number of mosquito eggs by treatment group:

Our model predicts that a control site will have 41.6% more mosquito eggs than the treated site (95% CI: [16.8, 71.7]). As we can see from the table and plot, the global effect of treatment group is statistically discernible ($Z = 3.540, p < 0.001$).

2.6.2 Effect of Time on Mosquito Intervention

To see if time affects the mosquito intervention's effect:

Figure 13: Treatment Group Estimated Marginal Means

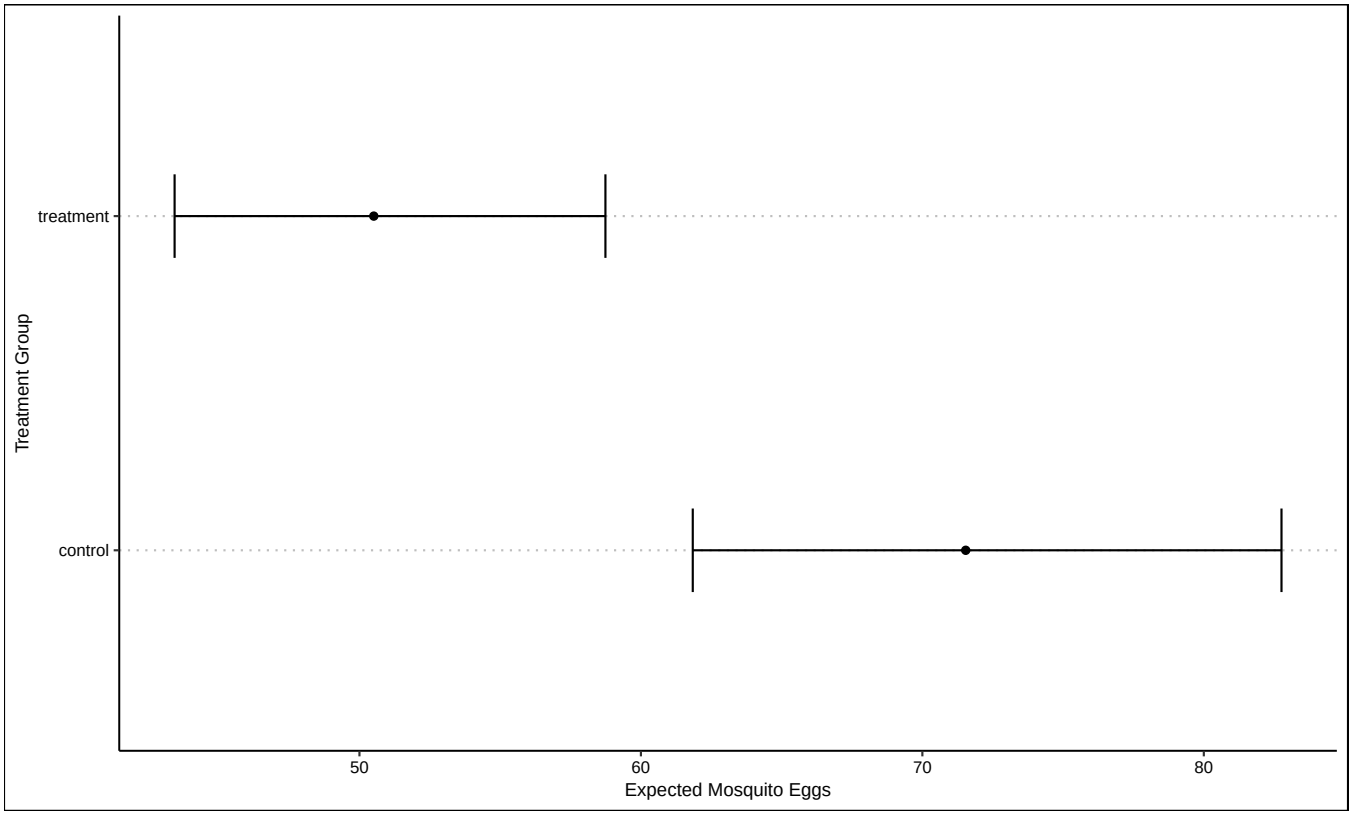


Table 5: Estimated Marginal Trend of Week by Treatment Group

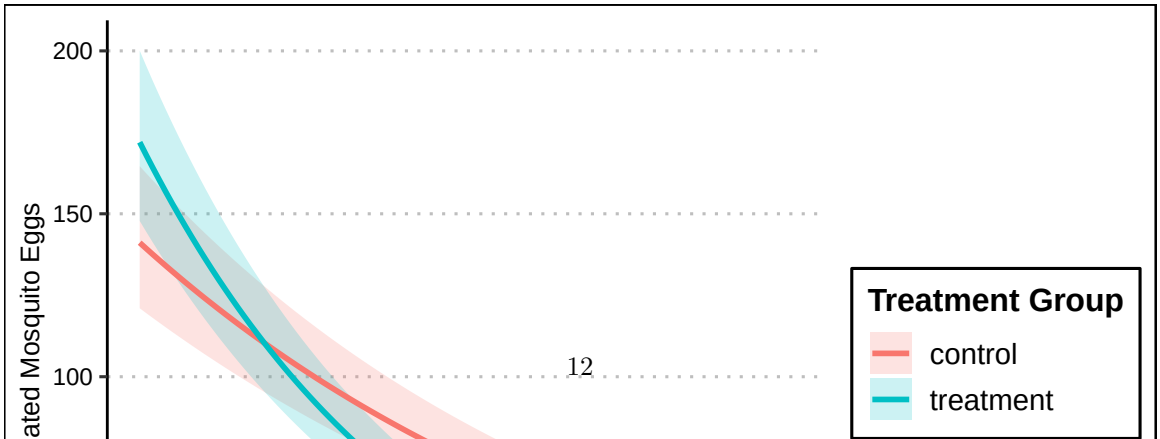
Treatment Group	Week Trend	SE	Z	p	adjusted-p	Percent Change	Lower CI	Upper CI
control	-0.116	0.007	-17.443	<0.001	<0.001	-10.976	-12.131	-9.805
treatment	-0.210	0.008	-27.320	<0.001	<0.001	-18.913	-20.123	-17.684

Both of the trends of week across groups are statistically discernible:

- Our model predicts that for every additional week, the number of mosquito eggs we'd see for the *control* group would decrease by 10.976% (95% CI: [9.805, 12.131]). This effect is statistically discernible ($Z = 17.443, p < 0.001$, adjusted-$p < 0.001$).
- Our model predicts that for every additional week, the number of mosquito eggs we'd see for the *treatment* group would decrease by 18.913% (95% CI: [17.684, 20.123]). This effect is statistically discernible ($Z = 27.320, p < 0.001$, adjusted-$p < 0.001$).

Visualizing those trends:

Figure 14: Estimated Marginal Trends of Week by Treatment Group



2.7 Conclusions

The results from our model are quite convincing. Before getting into those results, I should mention that the model is quite underdispersed, so perhaps another model would be better suited for this data.

Aside from that though, the treatment has very solid evidence for reducing the number of mosquito eggs. Just looking at the overall effect, the number of expected mosquito eggs is firmly lower in the treatment group (Figure 13). Often times, biological interventions can have quite small effects even when discernible, but an estimated 40% reduction in mosquito egss is quite striking.

The effect of time, specifically across treatment groups is also very interesting. When looking at the estimated marginal trends across treatment groups, the shapes of both curves appear to indicate an exponential decay in the expected mosquito eggs (Figure 14), just with slightly greater decay in the treatment group. This lends to the idea that the treatment group helps reduce the number of mosquito eggs, but not only for those sites with the treatment, but also across the whole study area. It'd be interesting to see if this pattern would continue if they kept measuring the eggs across even more weeks.

3 Disagreements With the Article

3.1 Results

Unfortunately, while the authors of this article and I come to similar conclusions about the effectiveness of the treatment, the way we went about assessing those differences were quite different, and make direct comparisons challenging.

A direct comparison I do think I can make is against this sentence in their results:

> " We first fitted a generalized linear mixed model (GLMM) that indicated the number of eggs per trap was influenced by the sampling area as well as the sampling week ($z = 78.05$, $P < 0.001$)"

I think their wording here is a bit strange. I believe they're referring to the reaction term, but I wouldn't necessarily say that I would interpret the coefficient from an interaction term in that way. If they're referring to the individual terms, why give one Z-statistic and p-value? If we assume they're referring to the interaction term, their statistics are much much larger than my model came up with (Table 3). When I went looking for answers in their code as to why this discrepancy is there, I noticed all of their code in their repository is in python, though they say in the article their analysis was done in *R*.

The second large claim they make in the results is:

> "Post hoc comparisons showed no significant differences in egg counts between the control and treatment areas in the 3 weeks of baseline egg counting undertaken before the intervention. The first sampling after the 8-week intervention was completed, showed significantly lower egg counts at all treatment areas compared with all control areas (Fig. 2a and Supplementary Table 3). This trend continued in the second week of egg counts after the intervention. Egg numbers at treatment areas started to merge with control area egg counts in weeks 3 and 4 after the cessation of the intervention (Supplementary Table 3, and Fig. 2a)."

This is a very different way than I analyzed their data. If I attempt to analyze the estimated marginal means at each of the weeks, instead of looking at the trend of weeks:

Table 7: Estimated Marginal Means of Treatment Group by Week (Post-hoc)

Group	Week	Estimate	SE	Lower CI	Upper CI
control	1	141.134	11.081	121.003	164.614
treatment	1	171.971	13.286	147.807	200.085
control	2	125.643	9.602	108.166	145.945
treatment	2	139.447	10.507	120.302	161.638
control	3	111.853	8.373	96.589	129.530
treatment	3	113.074	8.388	97.772	130.770
control	9	55.680	4.332	47.804	64.853
treatment	9	32.143	2.676	27.303	37.841

Table 7: Estimated Marginal Means of Treatment Group by Week (Post-hoc)

Group	Week	Estimate	SE	Lower CI	Upper CI
control	10	49.568	3.978	42.354	58.012
treatment	10	26.064	2.270	21.974	30.915
control	11	44.128	3.670	37.491	51.939
treatment	11	21.134	1.932	17.668	25.280
control	12	39.284	3.397	33.160	46.541
treatment	12	17.137	1.647	14.195	20.690

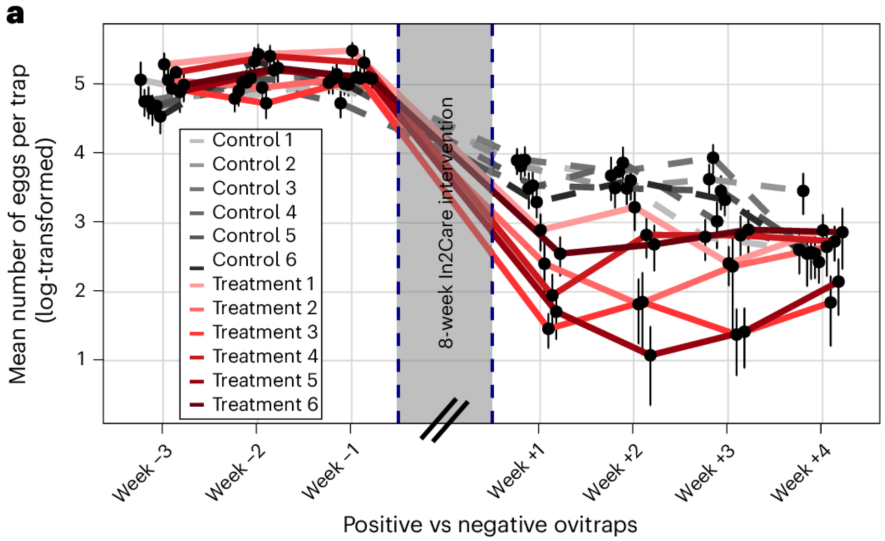
Table 8: Contrasts of Estimated Marginal Means of Treatment Gruop by Week (Post-hoc)

Contrast	Week	Esti-mate	SE	Z	p	adjusted-p	Percent Change	Lower CI	Upper CI
control - treatment	1	-0.198	0.105	-1.887	0.059	0.083	-17.931	-33.161	0.768
control - treatment	2	-0.104	0.101	-1.029	0.303	0.354	-9.899	-26.119	9.883
control - treatment	3	-0.011	0.099	-0.110	0.912	0.912	-1.079	-18.480	20.035
control - treatment	9	0.549	0.105	5.253	<0.001	<0.001	73.226	41.117	112.642
control - treatment	10	0.643	0.109	5.905	<0.001	<0.001	90.182	53.641	135.413
control - treatment	11	0.736	0.114	6.466	<0.001	<0.001	108.797	67.036	160.998
control - treatment	12	0.830	0.119	6.942	<0.001	<0.001	129.234	81.370	189.729

Looking at it this way, we can see that their first sentence makes sense. There are no discernible differences between the control and the treatment groups in the three weeks before the intervention, which makes sense. For the first two weeks after the intervention, weeks 9 & 10, we can now see that the difference between the two treatment groups is strongly discernible for both weeks. The part where I'm now confused, is when they say that the "Egg numbers at treatment areas started to merge with control area egg counts in weeks 3 and 4 after the cessation of the intervention." I'm not sure what led them to conclude that, given the estimated marginal means at weeks 3 & 4 after the intervention (weeks 11 & 12 in the model), are even more strongly different for the following weeks between the control and the treatment groups (Table 8). If you're wondering if their reference to figure 2a and Supplementary Table 3 clear this up, this is Figure 2a:

Figure 15: Paris et al.

From: Autodissemination stations suppress *Aedes notoscriptus* mosquitoes and reduce Buruli ulcer risk in urban Australia: a randomized controlled field trial



As you can see, not a figure which indicates statistics. Lastly, Supplementary Table 3 is simply the raw data for the study.

Sadly, that's where the discussion of the results for this analysis ends in their results section, so we're left on a bit of a cliffhanger for a lot of these results.

On one last interesting note from the results, the authors mention a period of low amounts of rain during the middle weeks of the study, which could explain a lot of the reduction in mosquito egg counts across both treatment groups. While this is helpful in adding context, I wonder why they didn't just include rainfall as a factor in the model if that data was available, instead of doing a separate analysis to test if rainfall alone was predictive of mosquito egg counts. The same could be said of temperature data.

3.2 Discussion

In the discussion, as well as the abstract, the authors bring up this statistic as a large piece of evidence for the effectiveness of the intervention:

> "The 70% reduction in egg counts in intervention areas compared to control areas we observed after the 8-week trial confirms the effectiveness of autodissemination systems for mosquito population reduction in urban settings."

I think the researchers might be slightly misinterpreting the data here. When they claim that there is a 70% reduction in egg counts in the intervention areas compared to the control groups, I think they really mean that the control groups have about 70% more mosquito eggs compared to the treatment groups, which is slightly different, and aligns with what my model predicts (Table 8). Also, if they wanted to have a more convincing statistic, I don't understand why they wouldn't go with the larger difference, which is at week 12, when the control groups have 129% more mosquito eggs than the treatment groups (Table 8).

So while the data is quite convincing for the efficacy of these interventions, these, what I would regard as misinterpretations or unclear results, unfortunatenly weaken their findings and message in my opinion.

4 Acknowledgement

I would like to thank the researchers who published this paper: Véronique Paris, Andrew H. Buultjens, Nicholas Bell, Thomas L. Schmidt, Wendy M. Bamberg, Michael T. Birrell, Finn Romanes, Catarina A. Antao, Maria Globan, Caroline Lavender, Katherine Bond, Sally Dougall, Jake A. Lacey, Yichen Zhai, Peihong Guo, Norelle L. Sherry, Katherine B. Gibney, Jue Tao Lim, Paul D. R. Johnson, Timothy P. Stinear & Ary A. Hoffmann. Transparency in data and code availability is hugely important for research ethics, and it makes my project possible.

5 AI Use

All of the guidance given by generative AI for this article can be found in this conversation.

6 Statistical Packages Used

- *tidyverse* for... everything it's *tidyverse* (Wickham et al. 2019)
- *ggplot2* for plotting (Wickham 2016)
- *ggthemes* for theme_clean() (Arnold 2024)
- *patchwork* for combined plots (Pedersen 2025)
- *glmmTMB* for inference of negative binomial GLMM model (Brooks et al. 2017)
- *performance* for model fit statistics (Lüdecke et al. 2021)

References

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